



Research Article

Processes accompanying ablation in thin-film and bulk chalcogenide glass As₂Se₃ under multipulse femtosecond laser irradiation



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Highlights

- Novel microstructures formed on As₂Se₃ thin films via IR femtosecond laser pulses.
- Dynamic waveguide model is proposed to explain the microstructure formation.

- Vapor-phase mechanism of As₂O₃ formation during laser ablation is proposed.
- Results are important for the development of IR photonics and high-power lasers.

Abstract

This paper studies the physicochemical processes accompanying ablation in thin-film and bulk arsenic selenide (As₂Se₃) under multipulse femtosecond laser irradiation in the visible and near-IR ranges. The morphological features of the irradiated samples and the chemical composition of the laser ablation products were studied using digital optical microscopy and Raman spectroscopy. For the first time, self-organized microstructures with a period on the order of the wavelength oriented parallel to the polarization vector were experimentally observed on As₂Se₃ thin films. Numerical modelling of the spatiotemporal distribution of As₂Se₃ complex refractive index allowed us to propose a formation mechanism for these structures based on the dynamic waveguide model. For bulk samples, the analysis of Raman spectra and assessments of the influence of the incubation effect and accumulative surface heating revealed that in the megahertz repetition rate range of femtosecond laser pulses, redox processes develop in the vapor phase, facilitating the formation of amorphous selenium (*a*-Se) and toxic arsenolite (As₂O₃) crystals in the deposited products. The results obtained are of interest for the fabrication of chalcogenide optical functional devices for IR photonics and the development of chalcogenide glass-based components in high-power laser systems.



Keywords

Arsenic selenide; Femtosecond laser modification; Femtosecond laser ablation; Periodic microstructure

1. Introduction

The increasing interest in chalcogenide glassy semiconductors (ChGSs) is primarily driven by their unique properties, as well as their low cost and ease of processing. Due to their ultrafast phase transitions, high refractive index, and transparency in the infrared region of the spectrum, glasses and thin films based on ChGSs serve as excellent materials for photonic applications. They are used as elements of infrared precision optics [1], find applications in memory devices [2], chemical sensors, and biosensors [3,4]. Optical components with surface microgratings based on ChGSs can significantly improve the compactness of optical systems [5]. Due to their high optical nonlinearity, ChGSs can be employed in the development of fiber laser amplifiers [6]. The expansion of ChGSs applications is possible by endowing them with new functional properties using fast and highly precise processing methods, among which femtosecond laser structural surface modification is particularly promising. By forming various types of polarization-dependent structures, it is possible to control the optical, mechanical, and chemical properties of material surfaces [7]. In particular, periodic structural changes on the surface of ChGS (As₂Se₃) irradiated by a femtosecond laser with an increasing number of pulses exhibit a pronounced structural color effect [8]. Moreover, due to the influence of various feedback mechanisms [9] that can arise under the action of femtosecond laser pulses on glassy semiconductors, a wide range of new effects with fundamental significance is possible.

Currently, a comprehensive theory of femtosecond laser surface modification is lacking. However, based on the Sipe-Drude theory [10], the work [11] predicts the possibility of forming various types of periodic surface microstructures, in particular, perpendicular to the polarization vector, parallel to the polarization vector, and cross microstructures with a period of the order of the radiation wavelength ($\Lambda \sim \lambda$). Moreover, the emergence of nanostructures with a period much smaller than the wavelength oriented both perpendicular and parallel to the polarization vector of the radiation is forecasted. At present, under femtosecond laser irradiation, predominantly low-spatial-frequency microstructures with $\Lambda \sim \lambda$ oriented perpendicular to the polarization vector and high-spatial-frequency nanostructures oriented parallel to the polarization vector, have been experimentally observed on the surface of ChGSs [12].

Low-spatial-frequency microstructures with $\Lambda \sim \lambda$ oriented parallel to the polarization vector, are much rarer and have been observed, in particular, on germanium sulfide-based chalcogenide glass (Ge-S) [13] and arsenic sulfide glass (As₂S₃) [14]. It should be noted that while the formation mechanism of surface structures with $\Lambda \sim \lambda$ oriented perpendicular to the polarization vector is based on the interference of the plasmon-polariton wave with the incident wave [15], the formation mechanism of microstructures oriented parallel to the polarization vector remains a subject of discussion. A promising representative of ChGSs

is arsenic selenide (As₂Se₃), which has a smaller band gap than arsenic sulfide (As₂S₃). However, the experimental demonstration of the formation of low-spatial-frequency microstructures parallel to the polarization vector on arsenic selenide films has not been achieved. Such microstructures are of interest for the fabrication of optical elements for IR photonics, particularly highly efficient polarization filters [16]. Due to the absence of a complete understanding of the formation of periodic surface structures on ChGs under femtosecond laser processing, industrial implementation of laser surface modification technology remains challenging.

When developing elements of high-power laser systems based on chalcogenide glasses, it is necessary to consider the high radiation intensities characteristic of femtosecond laser modes. The weak mechanical properties of ChGs may lead to material destruction, accompanied by chemical transformations of the glass. In particular, laser ablation of chalcogenide glasses containing arsenic can result in the release of toxic components, which is crucial to consider in the development of ChGs-based optical fibers to enhance their safety, including in medical applications such as laser surgery and tissue welding. The study [17] found that arsenolite (arsenic oxide, As₂O₃) crystals form on As₂S₃ and As₂Se₃ films during the formation of ablation lines in scanning mode at repetition rates of femtosecond laser pulses in the near-IR range up to 200kHz. In the mode of femtosecond laser surface modification of arsenic chalcogenide thin films in the visible range, such phase composition changes were not observed, although the structure formation threshold exceeded the film ablation threshold [12]. Moreover, with an increase in the number of laser pulses and their repetition rate, incubation and accumulative effects may manifest due to the low thermal diffusivity characteristic of ChGs [13,17,18].

Thus, the processes accompanying femtosecond laser ablation of chalcogenide glasses, including the formation of microstructures and phase transformations, remain poorly understood, and the existing data are fragmentary. Considering the promise of these materials for IR photonics and their use in high-power laser systems, a comprehensive study of these phenomena is an important area for both fundamental science and applied optics, ensuring increased reliability and functionality of chalcogenide optical components.

This paper presents the results of an experimental and theoretical study of the physicochemical processes accompanying ablation in thin-film and bulk arsenic selenide (As₂Se₃) under the influence of femtosecond laser pulses in the visible and near-IR ranges, depending on pulse energy, number, and repetition rate. The morphological features of irradiated samples and changes in their chemical and phase composition were studied using digital optical microscopy and Raman spectroscopy. The formation of previously unobserved low-spatial-frequency microstructures oriented parallel to the laser radiation polarization vector on

As₂Se₃ films was demonstrated. Based on numerical modelling of the spatiotemporal distribution of the complex refractive index and the heating dynamics of As₂Se₃ thin film, a mechanism for their formation was proposed. The evaluation of the incubation effect and accumulative surface heating of As₂Se₃ bulk chalcogenide glass revealed a vapor-phase mechanism of phase and chemical composition changes in the megahertz range of femtosecond pulse repetition rates.

2. Materials and methods

Thin-film and bulk arsenic selenide (As₂Se₃) were used as samples. As₂Se₃ thin films with a thickness of 500±5 nm were deposited by centrifugation of a solution of the corresponding materials onto a cover glass. The unpolished bulk samples with a thickness of ~3 mm were obtained by ultrasonic cutting from a piece of As₂Se₃ bulk glass, produced by direct fusion of materials according to the stoichiometric composition in quartz ampoules. The processing of bulk and thin-film samples was carried out in an air atmosphere at normal incidence of linearly polarized radiation from an ANTAUS-20W-20u/1 M femtosecond fiber laser (Avesta Project Ltd, Russia) with the wavelength of 515 nm and 1030 nm, the pulse duration of 224 fs, the pulse energy up to 20 μJ, and the pulse repetition rate up to 1 MHz. The laser radiation was focused onto the sample surface by a lens with a focal length of 45 mm. The diameter of the laser focal spot was determined using a well-known method [19], and the obtained values were in the range of ~20–40 μm (at the e^{-1} level) for wavelengths of 515 and 1030 nm. The sample was placed on a translation table. An automated system provided programmable positioning of the sample relative to the light beam and a multiple irradiation mode of the selected area with laser pulses. The microprocessing was monitored in real time using a CCD camera and illumination. The sample was placed in a plane orthogonal to the optical axis, and the crater array was recorded under different experimental conditions. The average power/energy of laser radiation was measured using a Gentec-EO SOLO2 meter (Gentec Electro-Optics, Inc., Canada). For the analysis and processing of images obtained using the Axio Imager A1m digital optical microscope (Carl Zeiss Microscopy GmbH, Germany), the ZEN lite 2 software (Carl Zeiss Microscopy GmbH, Germany) was used. Structural studies of the chemical composition were carried out using the Raman spectroscopy technique in the backscattering geometry using μ-Raman spectrometer Senterra (Bruker, Germany). The Raman scattering excitation was performed using the 785 nm wavelength solid state laser. The laser radiation power was 0.3 mW under the 10x objective with the numerical aperture of 0.25. Spectra were obtained in the 80–1545 cm⁻¹ spectral region using the diffraction grating of 1200 l/mm. The rectangular aperture size was 25x1000 μm. The accumulation time for recording was selected individually and ranged from 50 to 100 s. The number of repetitions was 2 items. Before and during the measurement, the calibration of the spectrometer was checked using the silicon peak 520.7 cm⁻¹.

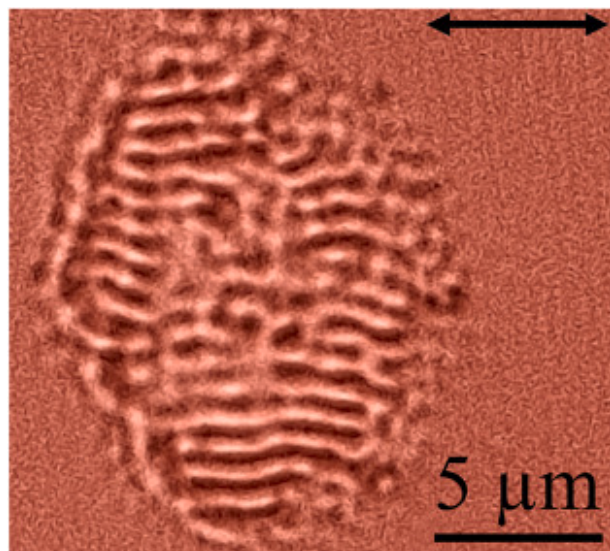
Estimated calculations of accumulative heating and numerical modelling of heating and photoexcitation of ChGS by a single femtosecond laser pulse were carried out in the Mathcad Prime 7.0 program (PTC, Inc, USA) and in the original program developed by the authors, respectively.

3. Experimental results and discussion

3.1. As₂Se₃ thin film

Irradiation of As₂Se₃ films with femtosecond laser pulses with the wavelength of 1030nm in sub-ablation and ablation modes allows us to determine the ablation thresholds using the method described in Ref. [19], based on the dependence of crater diameter squares on the natural logarithm of the incident laser pulse energy. The single-pulse laser ablation threshold of the As₂Se₃ film was $\sim 0.55 \text{ J/cm}^2$, and it was $\sim 0.53 \text{ J/cm}^2$ for $N=10$ pulses at a laser pulse repetition rate of $f=10 \text{ Hz}$. These values fall within the ablation threshold ranges for As₂S₃ and As₂Se₃ films determined in Ref. [17] in scanning modes under similar femtosecond laser irradiation parameters.

Further research is focused on identifying low-spatial-frequency ($\Lambda \sim \lambda$) periodic structures oriented parallel to the polarization vector. The analysis of two-dimensional crater arrays formed on the surface of As₂Se₃ thin film under irradiation with different numbers of femtosecond laser pulses N at various fluencies Q_0 at a specific repetition rate revealed modes in which periodic structures with stripes oriented along the polarization direction of the incident radiation form at the crater bottom. Their period determined by averaging data from different optical images of craters is close to the laser wavelength and is $\Lambda = 0.8 \pm 0.1 \mu\text{m}$. A typical appearance of low-spatial-frequency microstructures oriented parallel to the polarization vector obtained at the bottom of As₂Se₃ film crater at a fluence of $\sim 0.6 \text{ J/cm}^2$ and $N=2$ is shown in Fig. 1. The formation of periodic structures perpendicular to the polarization vector is observed locally at the crater periphery. Below the ablation threshold, the formation of low-spatial-frequency structures does not occur, nor does it occur at $N=1$ across the entire investigated fluence range. This fact significantly distinguishes laser processing of transparent materials from that of absorbing materials. In absorbing materials (e.g., metals, silicon), femtosecond laser surface modification is usually performed near the sub-ablation threshold [20]. In transparent materials (e.g., fused silica), the modification threshold exceeds the ablation threshold [21].



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Fig. 1. Typical optical image of a crater on the surface of As₂Se₃ thin film formed by irradiation with femtosecond laser pulses ($\lambda=1030\text{ nm}$, $\tau\sim 224\text{ fs}$, $N=2$, $f=10\text{ Hz}$, $Q_0=0.6\text{ J/cm}^2$). The black arrow shows the direction of the polarization vector.

As far as the authors are aware, low-spatial-frequency microstructures on the surface of As₂Se₃ thin films oriented parallel to the polarization vector have been obtained for the first time. Their appearance does not fit the classical model of laser-induced periodic surface structures (LIPSS) formation under laser irradiation which assumes the excitation of surface plasmon-polaritons followed by their interference with the incident radiation and predicts that microstructure stripes should be oriented perpendicular to the polarization vector of the incident radiation. This type of microstructuring was previously observed by the authors of [12] on As₂Se₃ thin films subjected to femtosecond laser processing, yielding microstructures with $\Lambda \sim \lambda$ perpendicular to the polarization vector. The formation mechanism of structures oriented parallel to the polarization direction requires separate consideration. For the first time, the original "waveguide" mechanism for the formation of such structures was proposed and implemented in Ref. [22]. At the interface between an absorbing semiconductor and a chemically active transparent liquid, laser-activated surface etching resulted in the formation of a gradient waveguide, where a waveguide mode was excited. Its interference with the incident radiation led to spatially modulated semiconductor etching and the formation of low-spatial-frequency structures parallel to the polarization vector. Waveguide structures in chalcogenide glasses were first obtained via

direct femtosecond laser writing in As₂S₃ thin films [23]. The authors noted that the change in refractive index correlated with photoinduced structural changes and was caused by variations in free electron concentration.

In [24], it was hypothesized that femtosecond laser irradiation of monocrystalline silicon forms a layered optical structure in the near-surface layers, potentially acting as a waveguide. The conversion of part of the incident radiation into waveguide modes, followed by its interference with the incident radiation, leads to the formation of low-spatial-frequency microstructures on the silicon surface oriented parallel to the polarization direction. It was suggested that external emission of excited electrons could play a role in creating the layered optical structure. Further studies within the framework of a more general theoretical model [25] showed a low intensity of electron emission, indicating that the nature of the layered optical structure requires further investigation. In Ref. [26], an explanation for the formation of periodic structures parallel to the polarization vector of laser radiation on silicon under multipulse femtosecond exposure was proposed, based on the development of thermocapillary instabilities in the melt pool. However, this explanation is unlikely to directly apply to the present study, as the authors considered large-scale structures with periods of $\Lambda \sim (2-3)\lambda$.

To determine the mechanism of low-spatial-frequency microstructure formation parallel to the polarization vector and to analyze in detail the heating and photoexcitation processes of the chalcogenide glassy semiconductor that may influence the microstructuring behavior of the material under femtosecond laser pulse exposure, we conducted numerical modelling based on the density-dependent two-temperature model (n-TTM). Since values for some critical parameters of As₂Se₃ required for comprehensive modelling are absent in the literature, we employed a simplified formulation of the n-TTM model [27], considering only the primary and most significant aspects of ChGS photoexcitation and heating by an ultrashort laser pulse while relying on approximate estimates for some material parameters. The heating and photoexcitation processes of the ChGS are described by a system of equations for the diffusion of photoexcited carriers (Eq. (1)), thermal conductivity for the electron and lattice subsystems (Eq. (2)), and an Eq. (3) describing the attenuation of incident radiation power density as it propagates into the material. The model takes into account the processes of two-photon interband absorption, free-carrier absorption, Auger recombination, carrier diffusion, electron-ion heat exchange, and thermal conductivity in the electron and lattice subsystems. In the absence of degeneracy in the electron subsystem [27], the system of equations with appropriate boundary conditions (Eq. (4)) to be solved takes the following form

$$\frac{\partial n_e}{\partial t} + \frac{\partial j_n}{\partial x} = \frac{\beta_2 q^2}{2\hbar\omega} - \gamma_{Au} n_e^3 \quad j_n = -D_e \left(\frac{\partial n_e}{\partial x} + \frac{n_e}{2T_e} \frac{\partial T_e}{\partial x} \right), \quad (1)$$

$$3n_e k_B \frac{\partial T_e}{\partial t} + (E_g + 3k_B T_e) \frac{\partial n_e}{\partial t} + \frac{\partial j_u}{\partial x} = \beta_2 q^2 + \alpha q - C_{ei} n_e (T_e - T_i) \quad (2)$$

$$(c_i + L_m \delta(T_i - T_m)) \frac{\partial T_i}{\partial t} - \frac{\partial}{\partial x} \left(k_i \frac{\partial T_i}{\partial x} \right) = C_{ei} n_e (T_e - T_i) \quad ,$$

$$j_u = (E_g + 4k_B T_e) j_n - k_e \frac{\partial T_e}{\partial x}$$

$$-\frac{\partial q}{\partial x} = \beta_2 q^2 + \alpha q, \quad (3)$$

$$\left. \frac{\partial n_e}{\partial x} \right|_{x=0} = \left. \frac{\partial n_e}{\partial x} \right|_{x=H} = 0 \quad \left. \frac{\partial T_e}{\partial x} \right|_{x=0} = \left. \frac{\partial T_e}{\partial x} \right|_{x=H} = 0 \quad \left. \frac{\partial T_i}{\partial x} \right|_{x=0} = \left. \frac{\partial T_i}{\partial x} \right|_{x=H} = 0, \quad (4)$$

$$q|_{x=0} = (1 - R) q_0 \quad n_e|_{t=0} = n_{e0} \quad T_e|_{t=0} = T_0 \quad T_i|_{t=0} = T_0$$

where n_e is the concentration of photoexcited carrier, T_e is the temperature of electron subsystem, T_i is the lattice temperature, q is the power density, D_e is the carrier diffusion coefficient, γ_{Au} is the Auger recombination coefficient, β_2 is the two-photon absorption coefficient, ω is the radiation frequency, $\hbar$ is the reduced Planck constant, k_B is the Boltzmann constant, k_e is the thermal conductivity of electrons, E_g is the band gap energy, C_{ei} is the parameter that determines the rate of electron-phonon relaxation, c_i is the heat capacity of the lattice, k_i is the lattice thermal conductivity, T_m is the melting temperature, L_m is the specific heat of fusion per unit volume, α is the free carrier absorption coefficient, R is the reflection coefficient, q_0 is the incident power density, H is the film thickness, n_{e0} is the initial carrier concentration, T_0 is the initial temperature.

The reflection coefficient R is calculated using the Fresnel formula with the Drude permittivity $\tilde{\epsilon}$

$$R = \frac{(n-1)^2 + k_p^2}{(n+1)^2 + k_p^2} \quad (5)$$

$$\tilde{\epsilon} = n_0^2 - \frac{\omega_p^2}{\omega^2 + \gamma^2} + i \frac{\gamma}{\omega} \frac{\omega_p^2}{\omega^2 + \gamma^2},$$

$$\omega_p = \sqrt{\frac{e^2 n_e}{\epsilon_0 m_{opt}}}$$

$$\alpha = \frac{2\omega k_p}{c}$$

where n , k_p are the real and imaginary parts of the complex refractive index, n_0 is the refractive index of an unexcited

semiconductor, ω_p is the plasma frequency, γ is the collision frequency, e is the electron charge, ϵ_0 is the electric constant, m_{opt} is the reduced optical mass of carriers, c is the speed of light in vacuum.

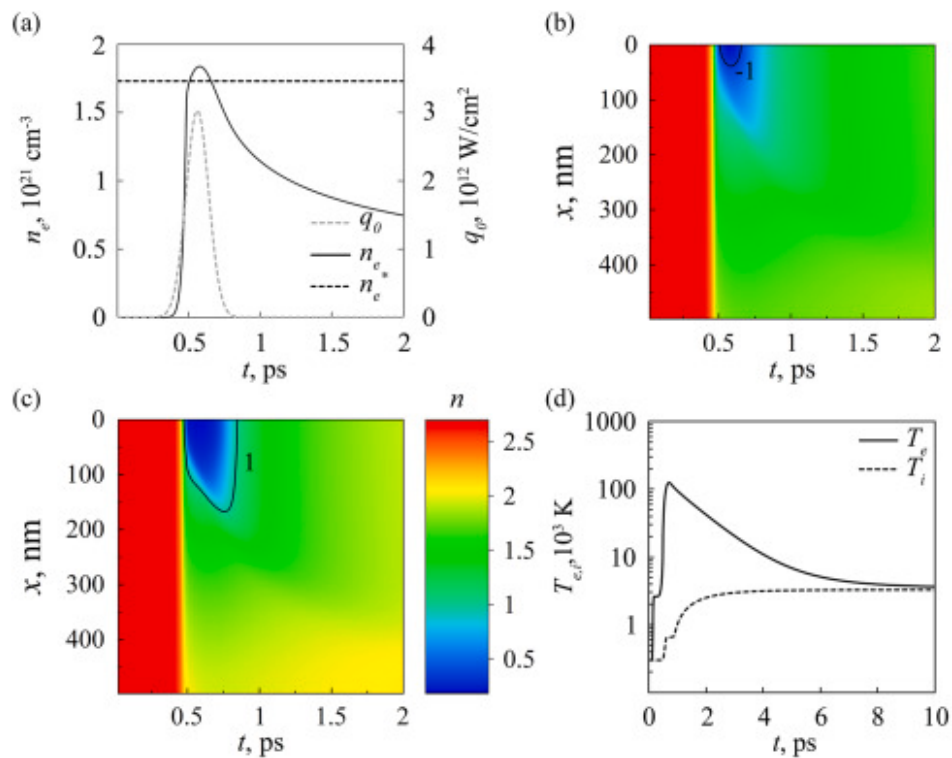
The diffusion coefficient and thermal conductivity of electrons are estimated based on the elementary kinetic theory of gases. The parameters used in the calculations are given in [Table 1](#).

Table 1. The parameters used in the calculations.

Parameter	Value	Comments
γ_{Au}	$4 \cdot 10^{-31} \text{ cm}^6/\text{s}$	The value for silicon from Ref. [27] was used.
β_2	$7 \cdot 10^{-9} \text{ cm/W}$ [28]	–
E_g	1.73 eV [29]	–
c_i	$1.57 \text{ J}/(\text{cm}^3 \cdot \text{K})$	Calculated by the formula $c_i = c \cdot \rho$, where $c = 0.33 \text{ J/g} \cdot \text{K}$ [30] is the heat capacity per unit mass, $\rho = 4.75 \text{ g/cm}^3$ [31] is the density.
k_i	$9 \cdot 10^{-3} \text{ W}/(\text{cm} \cdot \text{K})$ [32]	–
T_m	650 K [33]	–
L_m	$2 \cdot 10^3 \text{ J/cm}^3$	A value close to that for silicon was used [34].
C_{ei}	$4 \cdot 10^{-11} \text{ W/K}$	The value was chosen based on the consideration that $C_{ei} n_e \sim 10^{10} - 10^{11} \text{ W}/(\text{cm}^3 \cdot \text{K})$ [35].
m_{opt}	$0.2 m_e$	The value for silicon from Ref. [36] was used.
γ	$1 \cdot 10^{14} \text{ s}^{-1}$	Order of magnitude estimate. Large γ will lead to severe overheating.
n_0	2.7 [37]	–

[Fig. 2](#) shows the calculated time dependence of the photoexcited carrier concentration ([Fig. 2a](#)), the spatiotemporal distribution of the real part of the complex dielectric permittivity ([Fig. 2b](#)) and refractive index ([Fig. 2c](#)), as well as time dependences of the

electron and lattice temperatures at the surface of As₂Se₃ thin film (Fig. 2d).



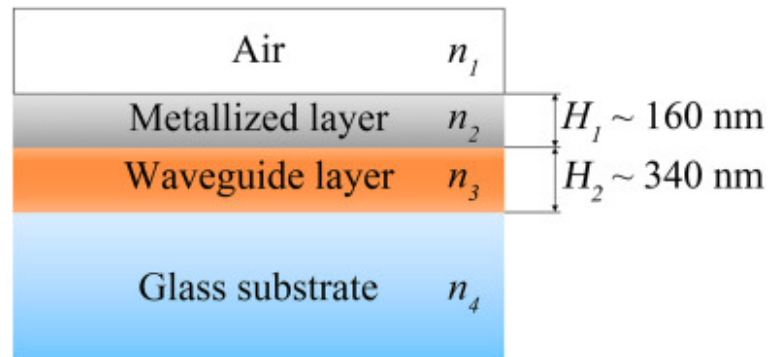
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Fig. 2. (a) Time dependence of free carrier concentration on the surface. The black dotted line marks the critical concentration n_{e*} for $\text{Re}(\epsilon)=-1$. Spatiotemporal distribution (b) of the real part of the complex dielectric permittivity and (c) the refractive index. (d) Time dependences of the electron and lattice temperatures on the surface of As₂Se₃ thin film. ($\lambda=1030$ nm, $\tau\sim 224$ fs, $Q_0=0.6$ J/cm²).

As can be seen from Fig. 2c, due to strong material photoexcitation during the femtosecond laser pulse, a metallized layer forms near the surface (i.e., a layer where the real part of the complex refractive index is less than 1). At the fluence of 0.6J/cm², the thickness of this layer is ~ 160 nm. Following this is a partially excited ChGS layer, with a refractive index varying from $n_1=1$ at the boundary with the metallized layer ($n_2<1$) to approximately $n_3\approx 2$ at the boundary with the glass substrate, which has the refractive index of $n_4=1.5$ (Fig. 3). The thickness of this partially excited layer is approximately 340nm or slightly less, considering

partial film ablation. Thus, a layered optical structure (dynamic waveguide) is formed during the action of the laser pulse, in which the thickness of the high-refractive-index layer (partially excited layer) exceeds the value of the minimum thickness of the flat layer required for the wavelength of 1030 nm, at which propagation of the waveguide mode is possible ($\sim 0.1\lambda$ [38]). The calculated time-dependent electron and lattice temperatures (Fig. 2d) indicate that in the picosecond range, the surface temperature of the film may exceed the boiling point of arsenic selenide [39], sufficient for enabling surface modification processes.



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Fig. 3. Schematic representation of a layered optical structure formed by the action of a femtosecond laser pulse on As₂Se₃ thin film ($n_3 > n_2, n_4$).

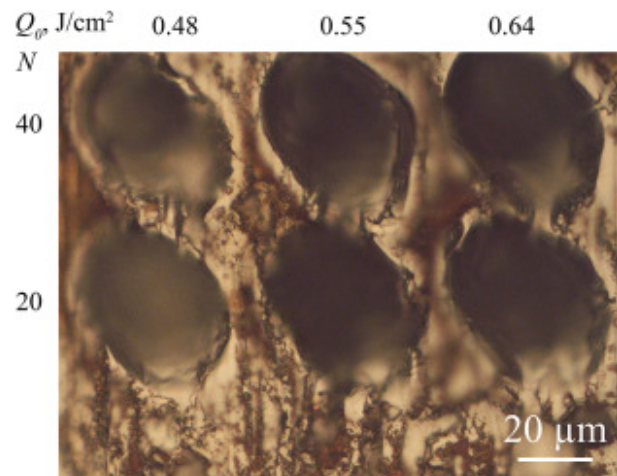
The proposed mechanism for the formation of periodic structures oriented parallel to the polarization vector is as follows. Under intense photoexcitation of the ChGS film by an ultrashort laser pulse, a layered optical structure is formed during the pulse action, which can act as a dynamic waveguide. In this case, the role of the waveguiding layer is played by a partially excited ChGS layer, confined between the metallized layer near the film surface and the glass substrate ($n_3 > n_2, n_4$). Due to the conversion of part of the incident radiation into a waveguide mode at inhomogeneities and its subsequent interference with the incident radiation, periodic structures oriented parallel to the polarization vector are formed. It should be noted that waveguide modes can exist only at high light intensities, and their excitation threshold exceeds the threshold for exciting plasmon-polariton waves [38]. In our case, the possibility of exciting a waveguide mode depends not only on the laser radiation intensity but also on the refractive index of the substrate. In Ref. [12], during the formation of hierarchical structures on As₂Se₃ films deposited on silicon

substrates, no development of microstructures with $\Lambda \sim \lambda$ oriented parallel to the polarization vector was observed. The presence of a silicon substrate with a higher refractive index ($n \sim 3.5$) likely disrupted the necessary condition for waveguide mode excitation. This fact indirectly supports the proposed "waveguide" mechanism of microstructuring.

As can be seen from [Fig. 1](#), the forming microstructures have a pronounced orientation, but their shape is not strictly symmetrical relative to the center of the laser spot. The waveguide mode, as a TE-type wave, determines the parallel orientation of the microstructures relative to the polarization vector, forming within the spatially modulated thermal field. The observed reduction in the period of the microstructures to $\sim 0.77\lambda$ may be caused not only by the propagation of the waveguide mode in a dielectric medium with a refractive index greater than 1 but also by additional transport of molten material induced by surface tension gradients (the Marangoni effect) [40], as well as partial evaporation accompanying the mass transfer process. The pronounced asymmetry and wave-like nature of the microstructures can be caused by the film surface microinhomogeneities, leading to local variations in the field intensity distribution, and the development of hydrodynamic instabilities due to the material evaporation in the laser beam center, where the radiation intensity is maximum.

3.2. As₂Se₃ bulk chalcogenide glass

The next series of experiments aimed to identify the chemical transformations of the chalcogenide glassy semiconductor accompanying its intense ablation by high-repetition-rate femtosecond laser pulses. Ablation crater arrays were recorded on the unpolished surface of a 3 mm thick As₂Se₃ chalcogenide glass under the influence of a series of femtosecond laser pulses with the wavelength of 515 nm at various pulse energies and repetition rates. This wavelength provides higher absorption of As₂Se₃ compared to the near-IR range [41], which facilitates effective local heating, evaporation of the substance, and initiation of phase transformations. This allows for a visual study of the processes accompanying laser ablation. The fragments of crater arrays recorded at a pulse repetition rate of 1 kHz are presented in [Fig. 4](#). It is evident that craters with smooth edges are formed, accompanied by the appearance of a rim and minor deposition of evaporation products.



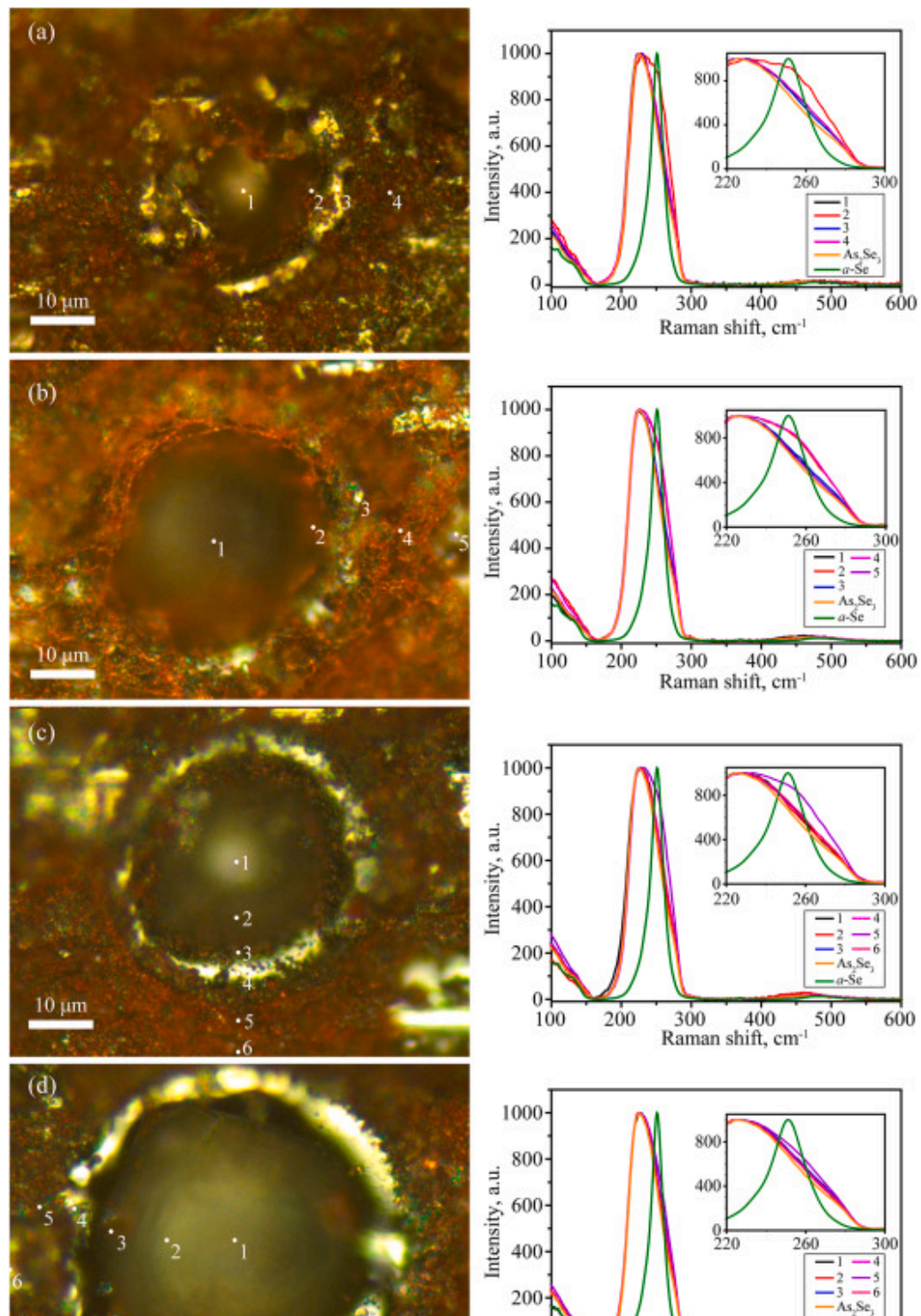
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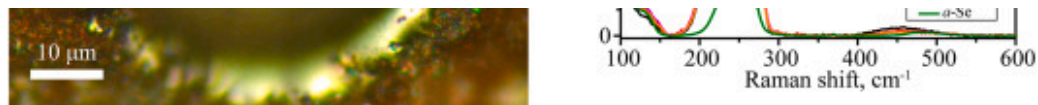
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Fig. 4. Optical image of a crater array fragment on the surface of As₂Se₃ bulk chalcogenide glass formed by irradiation with different numbers of femtosecond laser pulses N at a repetition rate $f=1$ kHz (vertically) and various fluencies Q_0 (horizontally) ($\lambda=515$ nm, $\tau\sim 224$ fs).

The single-pulse ablation threshold is ~ 0.1 J/cm², which is consistent in order of magnitude with literature data [41]. As the pulse number increases, an incubation effect appears, and the power-law dependence of the ablation threshold on the pulse number is given by $Q_N = Q_1 \cdot N^{S-1}$ [42]. The incubation coefficient S , calculated from the experimental dependence for As₂Se₃ at $f=1$ kHz, was found to be 0.94. For $N=1000$, the ablation threshold Q_{th} decreases to 0.05 J/cm². The incubation effect is based on structural changes on the surface occurring from pulse to pulse, which modify the absorptivity and lead to a reduction in the ablation threshold [21].

The situation changes dramatically when transitioning to high pulse repetition rates. Due to the low thermal diffusivity characteristic of chalcogenide glasses, an accumulation effect may occur. The formation of craters is accompanied by intense redeposition of laser ablation products (Fig. 5).





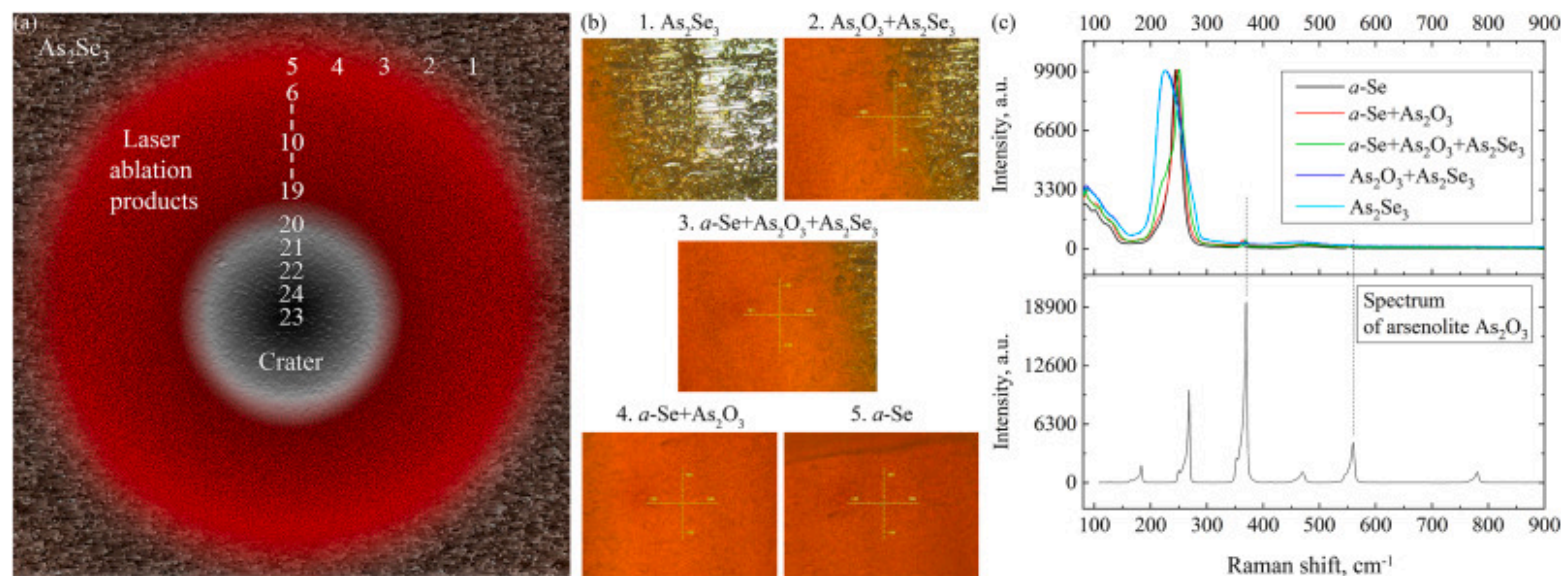
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Fig. 5. Optical images of craters on the surface of As₂Se₃ bulk chalcogenide glass obtained at 50x magnification with the measurement points of the Raman spectra (left) and the corresponding Raman spectra reflecting the chemical composition of the deposited products at the points (right). ($\lambda=515$ nm, $\tau\sim 224$ fs, $N=1000$, $f=1$ MHz, (a) $Q_0 \sim 5$ mJ/cm², (b) $Q_0 \sim 6.1$ mJ/cm², (c) $Q_0 \sim 6.3$ mJ/cm², (d) $Q_0 \sim 10.5$ mJ/cm²).

To determine the chemical composition of the ablation products deposited on the As₂Se₃ surface, Raman spectroscopy analysis was performed. Fig. 5 shows optical images of craters, obtained at 50x magnification, with bright spots in which the chemical composition of the deposited products was determined using Raman spectroscopy. Analysis of the normalized Raman spectra obtained in the center of the craters (point 1 in Fig. 5a–d), in the region of the melt rim (point 3 in Fig. 5a and b), point 4 in Fig. 5c and d) and near the craters (point 4 in Fig. 5a and b), points 5 and 6 in Fig. 5c and d), shows that similar structural features are observed for all craters formed by irradiation with femtosecond laser pulses at $N=1000$, $f=1$ MHz and $Q_0 < 11$ mJ/cm². Their main difference is a slightly higher intensity of the bands in the range of 240–275 cm^{−1} compared to the spectra of the unirradiated region of As₂Se₃. This indicates a slight increase in the number of bonds with selenium, which may be due to the presence of Se₈ rings and/or an amorphous selenium layer (*a*-Se).

Further studies are aimed at a detailed analysis of changes in the morphology and chemical composition of the deposited ablation products with an increase in the laser fluence by approximately two orders of magnitude. Fig. 6 shows: (a) a schematic diagram of the measurement points for Raman spectroscopy, (b) optical images of the irradiated sample areas at points 1–5, and the corresponding chemical composition of the deposited products obtained from Raman spectra (c). Points 1–5 correspond to measurements along a horizontal line toward the central axis of the region with deposited products. Points 6 to 24 show results along a vertical line toward the center of the ablation crater in the image plane.



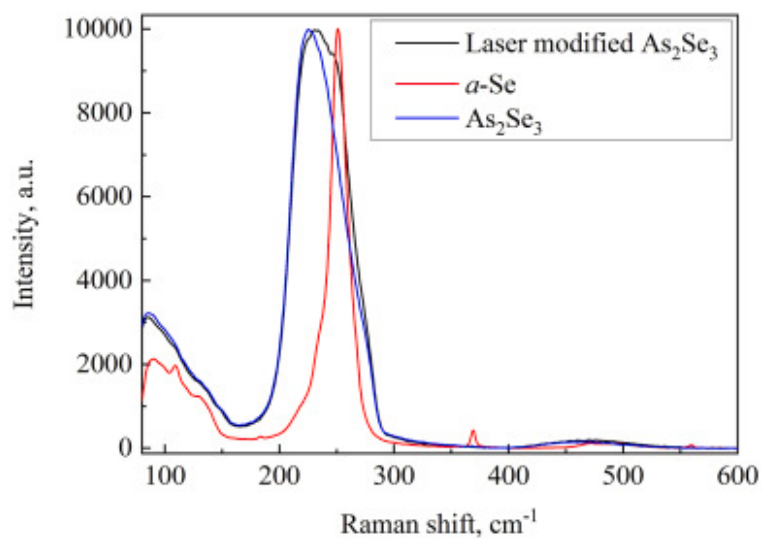
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Fig. 6. (a) Schematic diagram of the location of the Raman spectra measurement points. (b) Optical images of the irradiated sample regions at points 1–5 and the corresponding chemical composition of the deposited products ($\lambda=515\text{ nm}$, $\tau\sim 224\text{ fs}$, $Q_0\sim 0.7\text{ J/cm}^2$, $N=1000$, $f=1\text{ MHz}$). (c) Raman spectra for points 1–5 in the range of $80\text{--}900\text{ cm}^{-1}$, normalized to 10,000 rel. units, with vertical dotted lines showing the position of the As₂O₃ oxide peaks (top), and the reference Raman spectrum of the arsenolite mineral [43] (bottom).

A distinct pattern emerges when moving from the unirradiated area toward the deposition region. At the first point (Fig. 6b), only arsenic selenide (As₂Se₃) is detected (main peak at 225 cm^{-1}). At the second point, both arsenic selenide (As₂Se₃) and arsenic oxide (As₂O₃) contribute to the spectrum. At the third point, the spectrum closely resembles that of amorphous selenium (*a*-Se), which correlates with its characteristic reddish color. Additionally, this point exhibits the highest relative concentration of As₂O₃, with spectral peaks at 370 , 470 , and 558 cm^{-1} corresponding to the arsenolite mineral [43]. At the fourth point, the spectrum indicates contributions from amorphous selenium and arsenic oxide, while at the fifth point, the Raman spectrum closely resembles that of amorphous selenium (main broad band around 250 cm^{-1}). Further movement from point 6 to point 24 (Fig. 6a) reveals two distinct regions with qualitatively different compositions. The first region includes points 6 to 19, where amorphous

selenium (*a*-Se) and arsenic oxide (As₂O₃, in the form of arsenolite) are present at all points. The highest concentration of arsenic oxide (As₂O₃) is observed at points 10–14. From point 15 to point 19, the relative proportion of arsenic oxide gradually decreases. It should be noted that when analyzing the deposited products obtained as a result of femtosecond laser ablation at low laser fluencies ($Q_0 \sim 5\text{--}10\text{mJ/cm}^2$), it is not always possible to detect As₂O₃. The second region includes points 20 to 24, characterized by a predominant contribution from laser-modified As₂Se₃. The modification refers to changes in the ratio of structural units within the glass and/or the formation of new bonds compared to the original glass. At point 20, the spectrum shows the emergence of a shoulder around 250cm^{-1} . This spectral feature is close to the position of the most intense peak in amorphous selenium, but unlike the relatively narrow peak in amorphous selenium, the peak at point 20 is broader. This broadening may be attributed to the presence of Se₈ ring structures [44], whose vibrational frequencies fall within this range (Fig. 7). For points 21 to 24, the intensity of scattered light in the $230\text{--}280\text{cm}^{-1}$ range is significantly higher compared to unmodified glass, such as at point 1. The shoulder in this region is broader than in amorphous selenium, suggesting the presence of a greater number of structural units or selenium-related bonds, including Se₈ rings.



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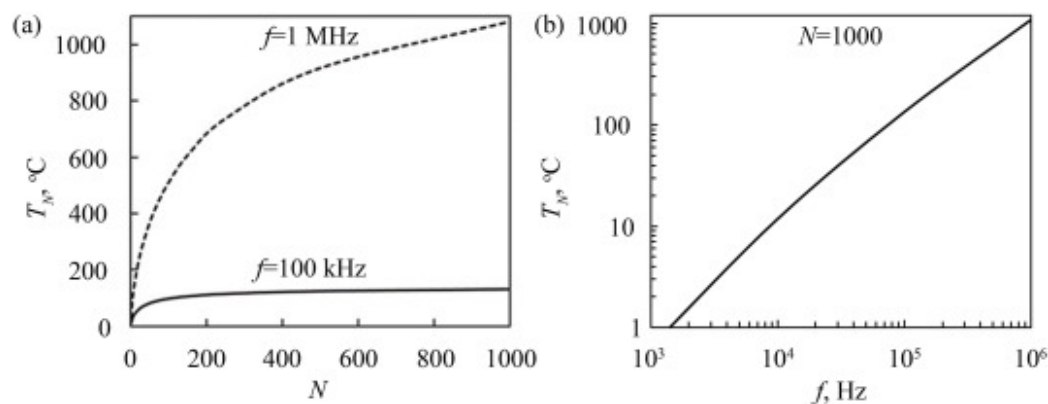
Fig. 7. Raman spectra for laser-modified As₂Se₃ (point 20 in Fig. 6a), *a*-Se (point 6 in Fig. 6a), and As₂Se₃ (point 1 in Fig. 6a).

The observed structural and chemical transformations may be influenced by the accumulative thermal effect accompanying laser exposure in the megahertz pulse repetition rate range, which, in turn, is the most efficient in terms of process productivity. Let us estimate the accumulative heating during multipulse laser irradiation using the results of [45]. Under the conditions of the present experiment, the residual temperature (relative to the initial temperature) at the beam center on the material surface is determined according to the following equation

$$T_N = \frac{(1-R)Q_0}{c_i} \sqrt{\frac{f}{\pi a}} \sum_{i=1}^N \frac{1}{\sqrt{i(\psi i + 1)}}, \quad (6)$$

where $R=0.25$ is the reflection coefficient of As₂Se₃ [46], $a = \frac{k}{c_i}$ is the thermal diffusivity coefficient of As₂Se₃, $k=3 \cdot 10^{-3} \text{ W}/(\text{cm} \cdot \text{K})$ is the thermal conductivity coefficient of As₂Se₃ [47], $\psi = \frac{4a}{fr_0^2}$ is the dimensionless parameter, $r_0=15 \text{ мкм}$ is the laser spot radius.

Fig. 8 shows the theoretical dependencies of the residual temperature T_N (relative to the initial temperature) on the As₂Se₃ surface at the center of the laser spot as a function of the number of femtosecond laser pulses N and their repetition rate f , calculated according to Eq. (6). As the pulse number increases, T_N reaches a steady-state value. At repetition rates below 1 kHz, accumulative heating is minimal but becomes significant as the pulse repetition rate approaches the megahertz range. At $f=1 \text{ MHz}$, accumulative heating can reach temperatures exceeding the boiling point of As₂Se₃ [39].



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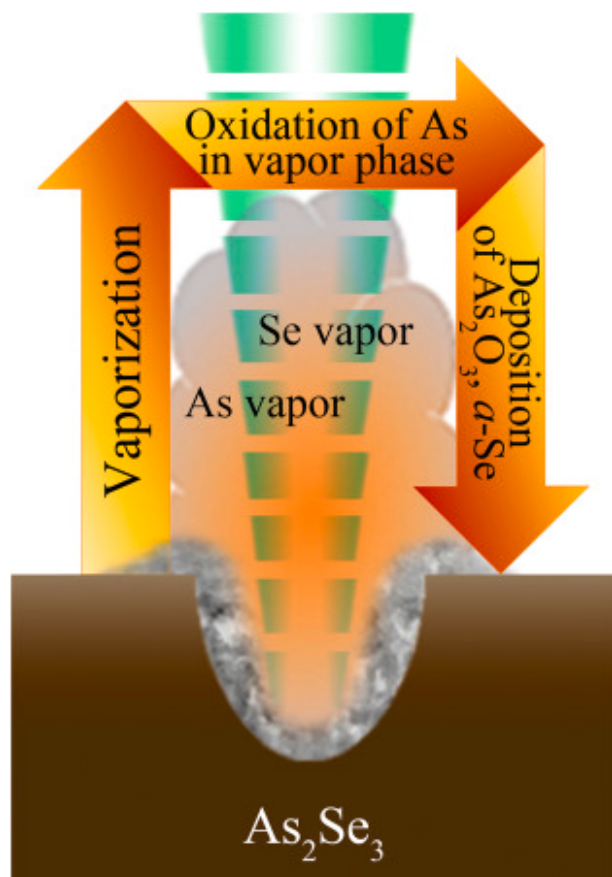
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Fig. 8. Theoretical dependences of the residual temperature T_N (relative to the initial temperature) in the center of the laser spot on the As₂Se₃ surface: (a) on the pulse number at repetition rates $f=100 \text{ kHz}$ and $f=1 \text{ MHz}$, (b) on the pulse repetition rate at

$N=1000$ ($\lambda=515\text{ nm}$, $\tau\sim 224\text{ fs}$, $d_0=30\mu\text{m}$, $Q_0=5\text{ mJ/cm}^2$).

Previously, in Ref. [17], the formation of arsenolite crystals was also observed on As₂Se₃ films when scanning femtosecond laser pulses at repetition rates of about 200kHz in through ablation modes. The authors proposed a solid-phase mechanism for its formation: according to the photodecomposition model [48], photon absorption with an energy exceeding the band gap energy leads to bond breaking and the formation of arsenic atoms, which can freely diffuse within the film and, in the presence of water vapor (acting as a catalyst), participate in the formation of As₂O₃.

Taken together, the results of digital optical microscopy and Raman spectroscopy, along with theoretical estimates of surface accumulative heating, allow us to conclude that the phase and chemical transformations in chalcogenide glass (As₂Se₃) during femtosecond laser ablation at megahertz repetition rates are thermal in nature, and the mechanism of arsenolite formation is vapor-phase (Fig. 9). Multipulse femtosecond laser exposure leads to the evaporation of arsenic selenide (As₂Se₃) and the formation of selenium and arsenic vapors. Since arsenic is the more volatile component, it spreads over greater distances from the irradiated area and oxidizes in the vapor phase upon interaction with atmospheric oxygen, resulting in the non-uniform deposition of arsenic oxide (As₂O₃) on the sample surface. Amorphous selenium primarily deposits closer to the crater. Within the crater itself, the modified As₂Se₃ exhibits an increased concentration of Se₈ ring structures compared to the original glass, contributing to a greater number of structural units and selenium bonds.



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Fig. 9. Schematic representation of the vapor-phase mechanism of arsenolite (As₂O₃) formation under the action of femtosecond laser pulses with a megahertz repetition rate on the As₂Se₃ chalcogenide glass surface.

4. Conclusion

On the surface of As₂Se₃ thin film exposed to near-IR femtosecond laser pulses above the ablation threshold, a previously unobserved type of microstructure with a period of $\sim \lambda$, oriented parallel to the polarization vector, was obtained. Based on a theoretical analysis of changes in the optical properties of photoexcited chalcogenide glass, a waveguide mechanism for their

formation was proposed. In the future, such microstructures could be utilized in the fabrication of optical elements for IR photonics, particularly for highly efficient polarization filters.

At pulse repetition rates of the order of 1 kHz, the incubation effect is observed in bulk As₂Se₃ glass, with an ablation threshold power-law dependence of $\sim N^{-1/16}$ on the pulse number. As the pulse repetition rate approaches the megahertz range, the accumulative thermal effect becomes significant. A combined analysis of digital optical microscopy and Raman spectroscopy data, as well as theoretical estimates of the surface accumulative heating of chalcogenide glass (As₂Se₃) under high-repetition-rate visible-range femtosecond laser pulses allowed us to propose and substantiate a vapor-phase mechanism for the formation of arsenic oxide (As₂O₃) around the crater. Within the crater region, a change in the ratio of structural units and the appearance of Se₈ ring structures were detected compared to the original glass.

The use of chalcogenide glasses with arsenic elements at high radiation intensities in the megahertz range of femtosecond laser pulse repetition rates requires increased safety precautions. Thus, the results obtained simultaneously contribute to the expansion of the functionality of As₂Se₃ in IR photonics and provide fundamental data for the development of reliable and robust components of high-power lasers.

CRedit authorship contribution statement

Alexandra Shamova: Writing – original draft, Visualization, Investigation, Formal analysis, Conceptualization. **Dmitry Polyakov:** Writing – review & editing, Methodology, Conceptualization. **Vladimir Golubkov:** Writing – review & editing, Investigation. **Dmitrii Pankin:** Writing – review & editing, Investigation, Formal analysis. **Alexander Shimko:** Writing – review & editing, Investigation, Formal analysis. **Galina Shandybina:** Writing – original draft, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] W. Wang, D. Qi, P. Lei, W. Shi, Z. Li, J. Zhang, W. Ho, H. Zheng
Chemical etching-assisted femtosecond laser multi-beam rapid preparation of As₂Se₃ microlens arrays
J. Manuf. Process., 120 (2024), pp. 460-466, [10.1016/j.jmapro.2024.04.077](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [2] N. Mehta
Advances in chalcogenide glasses (ChGs): past, present, and future applications
Advances in Glass Research, Springer International Publishing, Cham (2023)
[Google Scholar](#) ↗
- [3] A.A. Popescu, D. Savastru, M. Stafe, S. Miclos, L. Baschir, N.N. Puscas
Optical reflection of As₂Se₃-Au plasmonic planar structure for chemical sensors applications
UPB Sci. Bull. A: Appl. Math. Phys., 82 (2020), pp. 195-200
https://www.scientificbulletin.upb.ro/rev_docs_arhiva/rez582_711797.pdf ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [4] P. Lucas, M.A. Solis, D. Le Coq, C. Juncker, M.R. Riley, J. Collier, D.E. Boesewetter, C. Boussard-Plédel, B. Bureau
Infrared biosensors using hydrophobic chalcogenide fibers sensitized with live cells

Sensor. Actuator. B Chem., 119 (2006), pp. 355-362, [10.1016/j.snb.2005.12.033](https://doi.org/10.1016/j.snb.2005.12.033) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [5] M. Kang, B.M. Triplett, M.Y. Shalaginov, S. Deckoff-Jones, C. Blanco, M. Truman, E. Shirshneva-Vashchenko, J. Cook, Q. Du, T.S. Karnik, C.-C. Popescu, A. Zachariou, Y. Zhang, C.M. Schwarz, S. An, C. Fowler, H. Zhang, I. Divliansky, L.B. Glebov, M.C. Richardson, A.M. Agarwal, C. Rivero-Baleine, J. Hu, T. Gu, K.A. Richardson
Photochemically engineered large-area arsenic sulfide micro-gratings for hybrid diffractive–refractive infrared platforms
Adv. Photonics Res., 5 (2024), Article 2300241, [10.1002/adpr.202300241](https://doi.org/10.1002/adpr.202300241) ↗
[Google Scholar](#) ↗
- [6] J.-L. Adam, X. Zhang (Eds.), Chalcogenide Glasses, Preparation: Properties and Applications (first ed.), Woodhead Publishing Limited, Cambridge (2014)
[Google Scholar](#) ↗
- [7] J. Bonse, S.V. Kirner, J. Krüger
Laser-induced periodic surface structures (LIPSS)
K. Sugioka (Ed.), Handbook of Laser Micro- and Nano-engineering, Springer, Cham (2021), pp. 1-59, [10.1007/978-3-319-69537-2_17-2](https://doi.org/10.1007/978-3-319-69537-2_17-2) ↗
[Google Scholar](#) ↗
- [8] Z. Li, S. Shanguan, W. Shi, W. Wang, D. Qi, F. Chen, H. Zheng
Femtosecond laser induced periodic subwavelength nanohole arrays structure on As₂Se₃ glass surface
Ceram. Int., 51 (2024), pp. 8211-8218, [10.1016/j.ceramint.2024.12.302](https://doi.org/10.1016/j.ceramint.2024.12.302) ↗
[Google Scholar](#) ↗
- [9] J. Bonse, S. Höhm, S.V. Kirner, A. Rosenfeld, J. Krüger
Laser-induced periodic surface structures — a scientific evergreen
IEEE J. Sel. Top. Quant. Electron., 23 (2016), Article 9000615, [10.1109/jstqe.2016.2614183](https://doi.org/10.1109/jstqe.2016.2614183) ↗
[Google Scholar](#) ↗
- [10] J.E. Sipe, J.F. Young, J.S. Preston, H.M. Van Driel

Laser-induced periodic surface structure. I. Theory

Phys. Rev. B, 27 (1983), pp. 1141-1154, [10.1103/PhysRevB.27.1141](https://doi.org/10.1103/PhysRevB.27.1141) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [11] J.L. Déziel, J. Dumont, D. Gagnon, L.J. Dubé, S.H. Messaddeq, Y. Messaddeq
Toward the formation of crossed laser-induced periodic surface structures

J. Opt., 17 (2015), Article 075405, [10.1088/2040-8978/17/7/075405](https://doi.org/10.1088/2040-8978/17/7/075405) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [12] D. Shuleiko, S. Zaboltnov, O. Sokolovskaya, M. Poliakov, L. Volkova, T. Kunkel, E. Kuzmin, P. Danilov, S. Kudryashov, D. Pepelayev, S. Kozyukhin, L. Golovan, P. Kashkarov
Hierarchical surface structures and large-area nanoscale gratings in As₂S₃ and As₂Se₃ films irradiated with femtosecond laser pulses

Materials, 16 (2023), p. 4524, [10.3390/ma16134524](https://doi.org/10.3390/ma16134524) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [13] S.H. Messaddeq, R. Vallée, P. Soucy, M. Bernier, M. El-Amraoui, Y. Messaddeq
Self-organized periodic structures on Ge-S based chalcogenide glass induced by femtosecond laser irradiation

Opt. Express, 20 (2012), pp. 29882-29889, [10.1364/OE.20.029882](https://doi.org/10.1364/OE.20.029882) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [14] S.H. Messaddeq, A. Dumont, A. Douaud, M. El-Amraoui, Y. Messaddeq
Formation of cross-superposed LIPSSs on bulk chalcogenide glasses using fs-laser

Adv. Opt. Technol., 7 (2018), pp. 311-319, [10.1515/aot-2018-0031](https://doi.org/10.1515/aot-2018-0031) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [15] J. Bonse, S. Gräf
Maxwell meets Marangoni — a review of theories on laser-induced periodic surface structures

Laser Photon. Rev., 14 (2020), Article 2000215, [10.1002/lpor.202000215](https://doi.org/10.1002/lpor.202000215) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [16] Z.S. Liu, S. Tibuleac, D. Shin, P.P. Young, R. Magnusson
High-efficiency guided-mode resonance filter
Opt. Lett., 23 (1998), pp. 1556-1558, [10.1364/OL.23.001556](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [17] K.T. Paula, N.S. Dutta, J.M. Almeida, L.K. Nolasco, M.B. Andrade, C.B. Arnold, C.R. Mendonça
Femtosecond laser induced damage threshold incubation and oxidation in AS₂S₃ and As₂Se₃ thin films
Appl. Surf. Sci., 654 (2024), Article 159449, [10.1016/j.apsusc.2024.159449](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [18] P. Masselin, D. Le Coq, E. Bychkov
Refractive index variations induced by femtosecond laser direct writing in the bulk of As₂S₃ glass at high repetition rate
Opt. Mater., 33 (2011) (2011), pp. 872-876, [10.1016/j.optmat.2011.01.014](#) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [19] J.M. Liu
Simple technique for measurements of pulsed Gaussian-beam spot sizes
Opt. Lett., 7 (1982), pp. 196-198, [10.1364/OL.7.000196](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [20] J. Bonse, S. Baudach, J. Krüger, W. Kautek, M. Lenzner
Femtosecond laser ablation of silicon—modification thresholds and morphology
Appl. Phys. A, 74 (2002) (2002), pp. 19-25, [10.1007/s003390100893](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [21] S. Höhm, A. Rosenfeld, J. Krüger, J. Bonse
Femtosecond laser-induced periodic surface structures on silica
J. Appl. Phys., 112 (2012), Article 014901, [10.1063/1.4730902](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [22] V.A. Sychugov, T.V. Tulaikova
Waveguide effect during photoetching of semiconductors
Sov. J. Quant. Electron., 14 (1984), pp. 301-302, [10.1070/QE1984v014n03ABEH004885](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [23] A. Zoubir, M. Richardson, C. Rivero, A. Schulte, C. Lopez, K. Richardson, N. Hô, R. Vallée
Direct femtosecond laser writing of waveguides in As₂S₃ thin films
Opt. Lett., 29 (2004), pp. 748-750, [10.1364/OL.29.000748](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [24] G.A. Martsinovsky, G.D. Shandybina, Y.S. Dement'eva, R.V. Dyukin, S.V. Zobotnov, L.A. Golovan', P.K. Kashkarov
Generation of surface electromagnetic waves in semiconductors under the action of femtosecond laser pulses
Semiconductors, 43 (2009), pp. 1298-1304, [10.1134/S106378260910008X](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [25] D.S. Polyakov, E.B. Yakovlev
Ultrafast electron transfer through a silicon–vacuum interface induced by the action of an intense femtosecond laser pulse
J. Phys. D Appl. Phys., 53 (2020), Article 055305, [10.1088/1361-6463/ab533e](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [26] G.D. Tsibidis, C. Fotakis, E. Stratakis
From ripples to spikes: a hydrodynamical mechanism to interpret femtosecond laser-induced self-assembled structures
Phys. Rev. B, 92 (2015), Article 041405, [10.1103/PhysRevB.92.041405](#) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [27] A. Rämer, O. Osmani, B. Rethfeld
Laser damage in silicon: energy absorption, relaxation, and transport
J. Appl. Phys., 116 (2014), Article 053508, [10.1063/1.4891633](#) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [28] F. Sanchez, G. Boudebs, S. Cherukulappurath, H. Leblond, J. Troles, F. Smektala
Two-and three-photon nonlinear absorption in As₂Se₃ chalcogenide glass: theory and experiment
J. Nonlinear Opt. Phys. Mater., 13 (2004), pp. 7-16, [10.1142/S0218863504001724 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] D.C. Sati, R. Kumar, R.M. Mehra
Influence of thickness on optical properties of a: As₂Se₃ thin films
Turk. J. Phys., 30 (2006), pp. 519-527
<https://journals.tubitak.gov.tr/physics/vol30/iss6/6> ↗
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] A.J. Easteal, J.A. Wilder, R.K. Mohr, C.T. Moynihan
Heat capacity and structural relaxation of enthalpy in As₂Se₃ glass
J. Am. Ceram. Soc., 60 (1977), pp. 134-138, [10.1111/j.1151-2916.1977.tb15488.x ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] P. Knotek, P. Kutálek, E. Černošková, M. Vlček, L. Tichý
The density, nanohardness and some optical properties of As–S and As–Se chalcogenide bulk glasses and thin films
RSC Adv., 10 (2020), pp. 42744-42753, [10.1039/D0RA08939G ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] J. Lonergan, C. Smith, D. McClane, K. Richardson
Thermophysical properties and conduction mechanisms in As_xSe_{1-x} chalcogenide glasses ranging from x= 0.2 to 0.5
J. Appl. Phys., 120 (2016), Article 145101, [10.1063/1.4962446 ↗](#)
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] Z.U. Borisova

Glasses in binary systems

Glassy Semiconductors, Springer, Boston, MA (1981), pp. 37-168, [10.1007/978-1-4757-0851-6_3](https://doi.org/10.1007/978-1-4757-0851-6_3) ↗

[Google Scholar](#) ↗

- [34] S. Tao, B. Wu, Y. Zhou, Y. Gao
Thermal modeling and experimental study of infrared nanosecond laser ablation of silicon
J. Appl. Phys., 106 (2009), Article 123507, [10.1063/1.3271413](https://doi.org/10.1063/1.3271413) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [35] K. He, Y. Ren, Z. Dai, J. Zhang, X. Tu, L. Cheng, Z. Xin, L. Cai, Y. Ye
Theoretical and experimental study of ablation of fused silica by femtosecond laser bursts
Opt. Commun., 537 (2023), Article 129440, [10.1016/j.optcom.2023.129440](https://doi.org/10.1016/j.optcom.2023.129440) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [36] S.I. Ashitkov, A.V. Ovchinnikov, M.B. Agranat
Recombination of an electron-hole plasma in silicon under the action of femtosecond laser pulses
JETP Lett., 79 (2004), pp. 529-531, [10.1134/1.1787099](https://doi.org/10.1134/1.1787099) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [37] S. Joseph, S. Sarkar, J. Joseph
Grating-coupled surface plasmon-polariton sensing at a flat metal–analyte interface in a hybrid-configuration
ACS Appl. Mater. Interfaces, 12 (2020), pp. 46519-46529, [10.1021/acsami.0c12525](https://doi.org/10.1021/acsami.0c12525) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [38] M.N. Libenson
Laser-Induced Optical and Thermal Processes in Condensed Media and Their Mutual Influence
Nauka, Saint Petersburg (2007)
(in Russian)
[Google Scholar](#) ↗

- [39] E.H. Baker
Arsenic triselenide: boiling-point relation at elevated pressures
J. Chem. Soc. Dalton Trans. (1975), pp. 1589-1591, [10.1039/DT9750001589](https://doi.org/10.1039/DT9750001589) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [40] T. Kunkel, Y. Vorobyov, M. Smayev, P. Lazarenko, A. Kolobov, S. Kozyukhin
Self-organized structures in thin films of phase-change material upon femtosecond laser excitation: from periodic ordering to ablation
Appl. Surf. Sci., 624 (2023), Article 157122, [10.1016/j.apsusc.2023.157122](https://doi.org/10.1016/j.apsusc.2023.157122) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [41] Q. Li, D. Qi, X. Wang, X. Shen, R. Wang, K. Tanaka
Femto-and nano-second laser-induced damages in chalcogenide glasses
Jpn. J. Appl. Phys., 58 (2019), Article 080911, [10.7567/1347-4065/ab3368](https://doi.org/10.7567/1347-4065/ab3368) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [42] Y. Jee, M.F. Becker, R.M. Walser
Laser-induced damage on single-crystal metal surfaces
JOSA B, 5 (1988), pp. 648-659, [10.1364/JOSAB.5.000648](https://doi.org/10.1364/JOSAB.5.000648) ↗
[View in Scopus](#) ↗ [Google Scholar](#) ↗
- [43] Database the RRUFF™ Project
<https://rruff.info/arsenolite/display=default/R050383> ↗ (2025), Accessed 25th Mar 2025
[Google Scholar](#) ↗
- [44] V. Kovanda, M. Vlček, H. Jain
Structure of As–Se and As–P–Se glasses studied by Raman spectroscopy
J. Non-Cryst. Solids, 326 (2003), pp. 88-92, [10.1016/S0022-3093\(03\)00383-1](https://doi.org/10.1016/S0022-3093(03)00383-1) ↗
 [View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [45] D. Polyakov, G. Shandybina, A. Shamova

Analytical 3D modeling of accumulative heating under multipulse laser irradiation of inorganic materials and biological tissues

Therm. Sci. Eng. Prog., 31 (2022), Article 101284, [10.1016/j.tsep.2022.101284](https://doi.org/10.1016/j.tsep.2022.101284) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [46] V.V. Sobolev
Optical spectra of arsenic chalcogenide glasses over a wide energy range of fundamental absorption

Glass Phys. Chem., 28 (2002), pp. 399-416, [10.1023/A:1021719017568](https://doi.org/10.1023/A:1021719017568) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [47] R. Flasck, H.K. Rockstad
The thermal conductivity of some chalcogenide glasses

J. Non-Cryst. Solids, 12 (1973), pp. 353-356, [10.1016/0022-3093\(73\)90007-0](https://doi.org/10.1016/0022-3093(73)90007-0) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [48] J.S. Berkes, S.W. Ing Jr., W.J. Hillegas
Photodecomposition of amorphous As₂Se₃ and As₂S₃

J. Appl. Phys., 42 (1971), pp. 4908-4916, [10.1063/1.1659873](https://doi.org/10.1063/1.1659873) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Cited by (2)

Unraveling power-dependent ablation morphologies, phase transition dynamics, and oxidation mechanisms in femtosecond laser ablation of molybdenum: Experiment and multi-scale modeling

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